

Lab #:

Date:

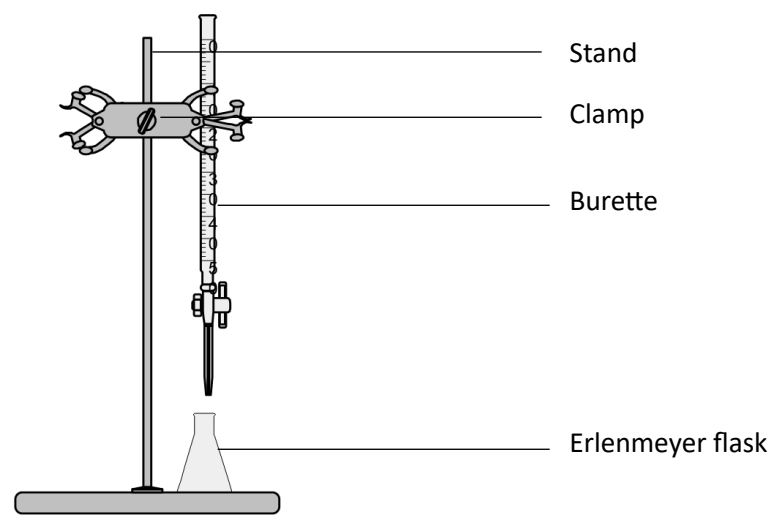
Aim: To determine the concentration of a solution of hydrochloric acid by titrating it against a standard solution of 0.10 mol dm^{-3} sodium hydroxide solution.

Title: Acid-base titration

Topic: Mole Concept

Apparatus/Materials:

Diagram:



Procedure:

1. Rinse the burette with distilled water followed by 5 mL of sodium hydroxide solution.
2. Fill the burette with the sodium hydroxide solution until it is just below the zero mark.
3. Rinse the pipette with distilled water followed by 5 mL of hydrochloric acid solution.
4. Rinse the Erlenmeyer flask with distilled water.

5. Pipette 25 mL of the hydrochloric acid solution into the Erlenmeyer flask.
6. Add two drops of phenolphthalein indicator to the hydrochloric acid in the Erlenmeyer flask and swirl for 15 seconds.
7. Record the initial burette reading.
8. Place the Erlenmeyer flask on a white tile below the burette and titrate the hydrochloric acid solution against the sodium hydroxide solution until there is a permanent colour change.
9. Record the final burette reading.
10. Repeat the experiment until concordant values are obtained.

Results:

Trial	1	2	3
Initial volume/mL			
Final volume/mL			
Titre volume/mL			

Calculations:

1. Determine the average titre volume used.
2. Write a balanced chemical equation for the reaction between hydrochloric acid and sodium hydroxide.
3. Determine the number of moles of sodium hydroxide present in the average titre volume.
4. Using the mole ratio from the balanced chemical equation, determine the number of moles of hydrochloric acid used in the pipetted volume (25 mL).

5. Determine the number of moles of hydrochloric acid present in 1 dm³ of solution.

Discussion:

1. Define acid-base titration.
2. Identify the standard solution and the analyte used in the titration.
3. Why was an indicator used in the titration? Explain.
4. At what point during the titration was there a colour change?

Sources of Error/Limitations:

- List two sources of error or limitations.

Precautions:

- List two precautions.

Conclusion:

- Based on the aim and results obtained, write a suitable conclusion.